

Michael W. Robbins

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PROFESSIONAL SUMMARY

- Ph.D. statistician with postdoctoral training, academic experience as a professor, and twelve years with the RAND Corporation, whose work blends methodological innovation with collaborative applied research and is supported by a strong record of extramural funding, project leadership, and software development.
- Authored approximately 100 peer-reviewed publications and have been cited over 1500 times.

RECENT EXPERIENCE

RAND Corporation

Pittsburgh, PA

Senior Statistician

2021– ...

Statistician

2015–2021

Associate Statistician

2013–2015

- Led ten research projects with a total value of over \$10 million as principal or co-principal investigator, including grants from the NSF, NIH, and various foundations, and contributed as the statistical lead or other role on over 100 other projects.
- Secured funding through proposals, formed and managed interdisciplinary research teams, oversaw execution of the work, and led dissemination of policy findings to technical and non-technical audiences including industry experts, project sponsors, and the general public.
- Developed innovative statistical methodologies in a wide variety of areas including causal inference, missing data analysis, sample design, survey weighting, nonprobability methods, and time series analysis.
- Performed collaborative research across several domestic policy areas (such as public health, education, criminology, and drug policy) and for military and defense sponsors, including the Department of Homeland Security (CBP, FEMA, OHSS, USCG, TSA), Department of Defense, Navy, and Army.
- Established an international reputation for expertise in survey methodologies, having performed sample design, weighting, and imputation for dozens of surveys.
- Evaluated 20+ interventions and programs using experimental & quasi-experimental causal inference tools.
- Synthesized complex analytic findings into actionable policy recommendations.
- Produced multiple R packages that are available for use on CRAN.

PROJECTS LED (SELECTED)

A Robust Design for Covert Testing of TSA Passenger Screening

2019–2025

- Designed a strategy for determining when and where the Transportation Security Administration should implement covert testing for airport screening given uncertain risk profiles.
- Demonstrated the positive impact of specific training programs on security officer performance.
- Benchmarked TSA's covert testing programs against industry standards, illustrating that TSA is a standard bearer in quality and rigor of covert testing while noting specific areas where improvements could be made.

A Multi-Phase Survey Strategy for Obtaining Representativeness of Big Data

2019–2024

- Created a method for estimating demographic characteristics of social media users based on their posts.
- Pioneered procedures for using generative AI to gauge sentiment expressed within sections of text.
- Administered a general population survey and a survey of Twitter users.
- Developed real-time estimates of nationwide sentiment of political candidates using only opinions expressed on Twitter by weighting a universe of Twitter users.

Data Fusion for Predicting Long-Term Program Impacts

2019–2022

- Developed a method that fuses multiple datasets to enable statistical estimation of the long-term effects of an intervention using short-term data.
- Used the method to estimate the effect of winning the Oregon Health Insurance Lottery on mortality.

Expand the American Educator Panels

2016–2019

- Sampled and recruited 30,000+ teachers and 13,000+ school leaders to join RAND's educator survey panels.
- Designed and performed scientific comparisons of strategies for recruitment.
- Developed algorithms for weighting survey data collected from the panels.

ADDITIONAL EXPERIENCE

Pardee RAND Graduate School

Santa Monica, CA

Professor

2017– ...

- Mentored graduate students (and junior analysts) on research projects.
- Led independent studies for graduate students and published scientific papers led by them.

University of Missouri, Columbia (Dept. of Statistics)

Columbia, MO

Assistant Professor

2011–2013

- Performed academic research and published in scientific journals.
- Taught undergraduate and graduate-level statistics courses.
- Performed departmental and professional service.
- Presented at national and international professional conferences.

National Institute of Statistical Sciences

Research Triangle Park, NC

Postdoctoral Fellow

2009–2011

- Managed a research project funded by the US Department of Agriculture and therein developed an algorithm for imputation of missing data in a large, complex, and high dimensional survey (the Agricultural Resource Management Survey)—the resulting procedure has been operationalized by the USDA.
- Collaborated with academic faculty, graduate students, and government statisticians and economists.
- Demonstrated the large improvement in performance of the new algorithm over the previous procedure.
- Published four scientific papers from the work.

EDUCATION

Clemson University

Clemson, SC

Ph.D. in Mathematical Sciences

- Thesis title: *Change-Point Analysis: Asymptotic Theory and Applications*.
- Developed an expertise in time series analysis, including a focus on structural break detection.

Master of Science in Mathematical Sciences

Duke University

Durham, NC

Bachelor of Science in Mathematics

Bachelor of Science in Economics

SKILLS

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|-------------------------|--------------------------|--------------------------------|
| • Project Management | • Machine Learning | • Numerical Optimization |
| • Software Development | • Program Evaluation | • Experimental Design |
| • Causal Inference | • Time Series Analysis | • Randomized Controlled Trials |
| • Missing Data Analysis | • Demographic Analysis | • Observational Studies |
| • Survey Methodology | • Epidemiology | • Econometrics |
| • Generative AI | • Operations Research | • Teaching |
| • Neural Networks | • Monte Carlo Simulation | • Mentoring |

SOFTWARE

- | | | | |
|----------|---------|---------|-----------------------------------|
| • R | • SAS | • Stata | • Microsoft Office |
| • Python | • C/C++ | • HTML | • L ^A T _E X |

AWARDS

- RAND Spotlight Award for Innovation (2024)

PRODUCTS

- Peer reviewed publications—see personal website or Google Scholar for lists.
- R package microsynth for disaggregated synthetic control estimation.
- R package gerbil for efficient joint imputation of general data.